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Automation In INVENTORY MANAGEMENT

- **Machine learning can help manage inventory far more efficiently by constantly tracking existing stock in the warehouse, its expiration date and forecasting future demands**

Manufacturers, e-commerce platforms and retailers need to store their inventories, which involve warehousing costs. Storing unnecessary stock can sink cash reserves when cash liquidity can be utilised for business growth. Whereas, insufficient stock can result in losing customers or possibly bring the whole production to a standstill until raw materials are available. Traditionally, businesses relied entirely on anticipation for forecasting stock demand.

Traditional inventory management techniques involved a high possibility of human errors and required a large workforce. Managing stock on paper or Excel sheets can be a solution for small stores, but big warehouses need better solutions. With the advent of enterprise resource planning (ERP) tools, businesses can track available items in stock and order when needed. However, how much they need to order is still an issue.

Internet of Things (IoT) technologies are creating an explosion of data, which can fuel machine learning (ML) algorithms. These algorithms can help manage inventory far more efficiently by constantly tracking existing stock in the warehouse, its expiration date and forecasting future demands. Using an artificial intelligence (AI) solution with ML algorithms for predictive analysis reduces the risk of human error. It can forecast future demands accurately, too.

AI is smart enough to timely analyse the requirements and place orders on behalf of employees, cutting down manual work and chances of human errors. AI-powered robots can now scan the shelves to check stock availability and replenish the stock without any human involvement. AI and ML can significantly reduce warehousing

costs. These technologies also reduce delivery or shopping times, which can result in positive customer reviews.

Significant reduction in stock on hand using AI solutions. An AI-based inventory management solution provider, Remi AI, developed and deployed such a platform in a Fortune 100 company. The large wholesale distributor with more than five warehouses in both the US and Canada was looking for ways to improve their distribution supply chain. This intelligent inventory platform was utilised to provide both demand forecasting and auto-replenishment solutions across its supply chain.

During the standard feasibility testing period, it was found that Remi AI's suite of demand-forecasting methods was accurate across 92 per cent of the client's product range and was set up to run at a stock keeping unit/warehouse/daily level.

AI-powered robots assisting Amazon to increase efficiency. There are currently more than 100,000 AI-powered robots working throughout Amazon's global fulfilment centres. These fulfilment centres are nearly the size of 28 football fields. As per Amazon's blog, having the robots wheel the inventory directly to fulfilment centres to retrieve ordered products saves significant time. Working in a technology-rich environment alongside robots also enables associates to be able to focus on more challenging tasks.

The robots open more space for inventory, making smaller buildings feasible and enabling faster shipping times and better prices for customers. These can lift weight that is way beyond the capacity of human beings.

Walmart using AI and ML for forecasting inventory. With AI and ML, Walmart watches which food items

will sell better. Predictive analysis and remote sensors tell distributors the slightest of details such as when a fridge needs restocking soda or when a coffee vending machine needs topping up. The company has also deployed shelves-scanning robots to check inventory, prices and misplaced items. These AI robots profoundly increase the efficiency and accuracy of the inventory management process.

Pick-up towers is another technologically-advanced solution deployed at Walmart stores to reduce long queues. Consumers can simply order through their phone and be notified when their order is complete. They need to visit the store and scan the QR code to get their package of goods from the conveyor belt. This technology has reduced the consumers' in-store time to a few minutes from hours.

If one aims at forecasting stock demands, one must check the nature of products. In case of a giant warehouse with numerous products, each product with a different nature of demand might not be the perfect scenario for deployment of AI technology for predictions. AI solutions are most ideal if the items have similar predictive natures and there is enough data available related to the demand and supply of products to run ML algorithms.

Technology is improving every aspect of business, and inventory management has a huge scope in terms of technological advancements. AI combined with ML and robotics can revolutionise the operations in inventory management. With automation, human intelligence can be removed from repetitive, manual and time-consuming tasks, to be utilised for customer engagement. **EFY**

—Deeksha Sharma